

# Build-Bench Challenge

## Autonomous LLM Agents for Cross-Architecture Package Repair

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# A working package can fail after migration

Cross-architecture repair connects source code, dependencies and packaging.



## Maintainer effort

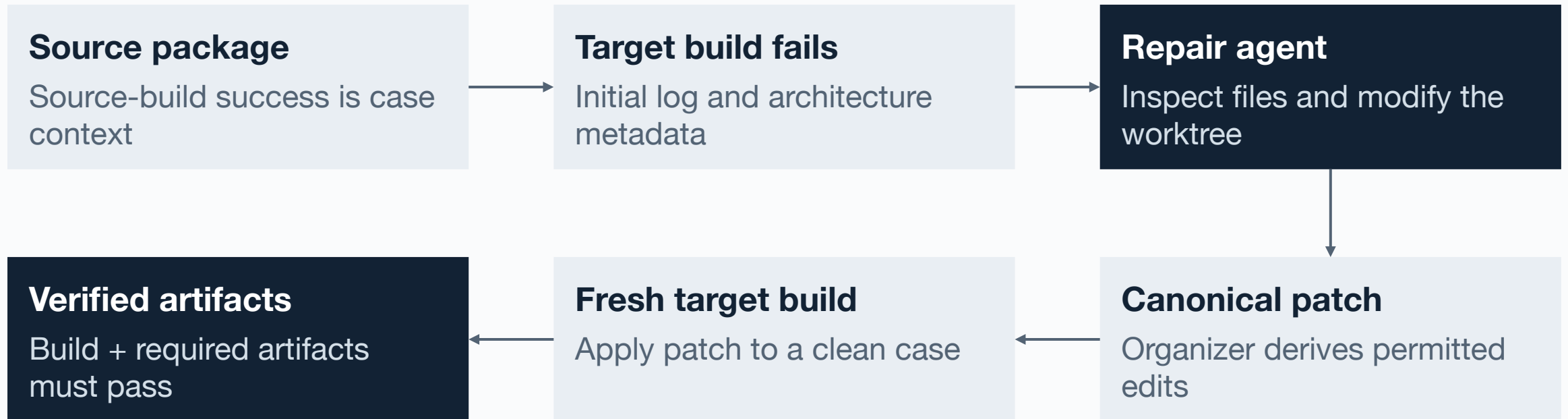
Trace the failure across unfamiliar logs, source and build files.

## Our objective

Automate diagnosis and repair.  
Confirm the result in a clean build.

# A clean target build determines success

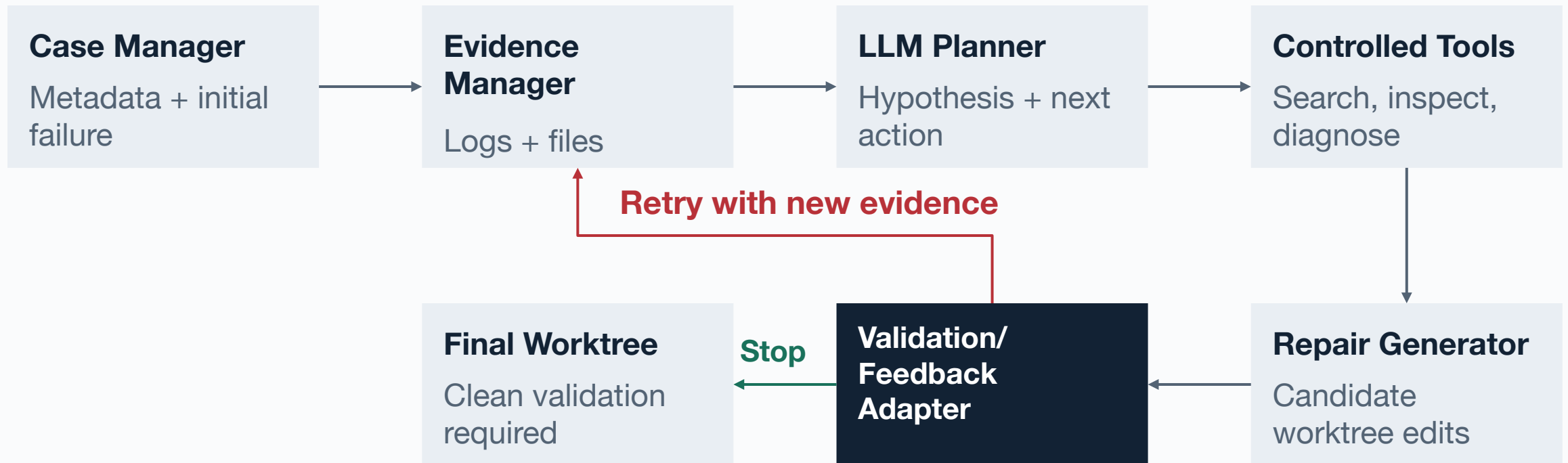
The organizer derives the patch and checks the resulting package artifacts.



**Public: x86\_64 and ARM64, both directions. Competition: also RISC-V**

# Proposed repair agent architecture

In-loop feedback depends on the interface. Final clean-build validation is required.



**History/Metrics records outcomes. Budget Controls enforce resource limits.**

# Each attempt tests a specific hypothesis

- 01**    **Observe**
- |
- 02**    **Diagnose**
- |
- 03**    **Gather evidence**
- |
- 04**    **Repair**
- |
- 05**    **Validate**

## **Bounded context**

One libyuv log is 2.37 GB.  
Select diagnostic windows and files.

## **Memory between attempts**

Keep hypotheses, diffs and failures.  
Stop repeated unsuccessful edits.

## **Initial research caps**

5 candidates / 30 min / 30k tokens

Proposed caps, subject to stricter organizer limits. Success or repetition can stop earlier.

# We will compare verified repairs and cost

## PRIMARY METRIC

# Verified Build Success Rate

Verified repairs  $\div$  all cases in the frozen evaluation set

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## Fair comparison

Same cases, model and budget ceilings  
Official baseline where reproducible  
Failures remain in the denominator

## Cost and failure records

Wall time and model tokens  
Repair attempts and tool calls  
Direction and failure breakdowns

# Semester milestones have checkable outcomes

October 21

## Demo 2

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Runnable LLM prototype  
10 fixed public cases  
Baseline + per-case records

November 16

## Demo 3

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Same 10 cases, 3 runs  
+1 mean repair, OR  
20% fewer model tokens  
preserving repaired cases

December 9

## Final

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40-case evaluation  
30 held out from tuning  
Reproducible agent,  
commands and results

**Competition freeze: November 13, before Demo 3. Qualification is conditional.**

Evaluation targets. Repeat all 10 cases, hold budgets fixed and report every regression.

# Organizer's hello workflow reproduced

ORGANIZER EXAMPLE, NO LLM

**Initial failure → patch → clean success**

**2 verified RPM artifacts**

**9 s final build**

Organizer's example only. This is not a repair result from our agent.

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## Hardest challenge

Choose useful evidence and revise diagnoses within a fixed budget.

## Immediate next steps

Plan AWS environments and pin configs.  
Reproduce a Debian case and confirm APIs.